

# SE205180\_TrانNguyenVietDung\_Lab02

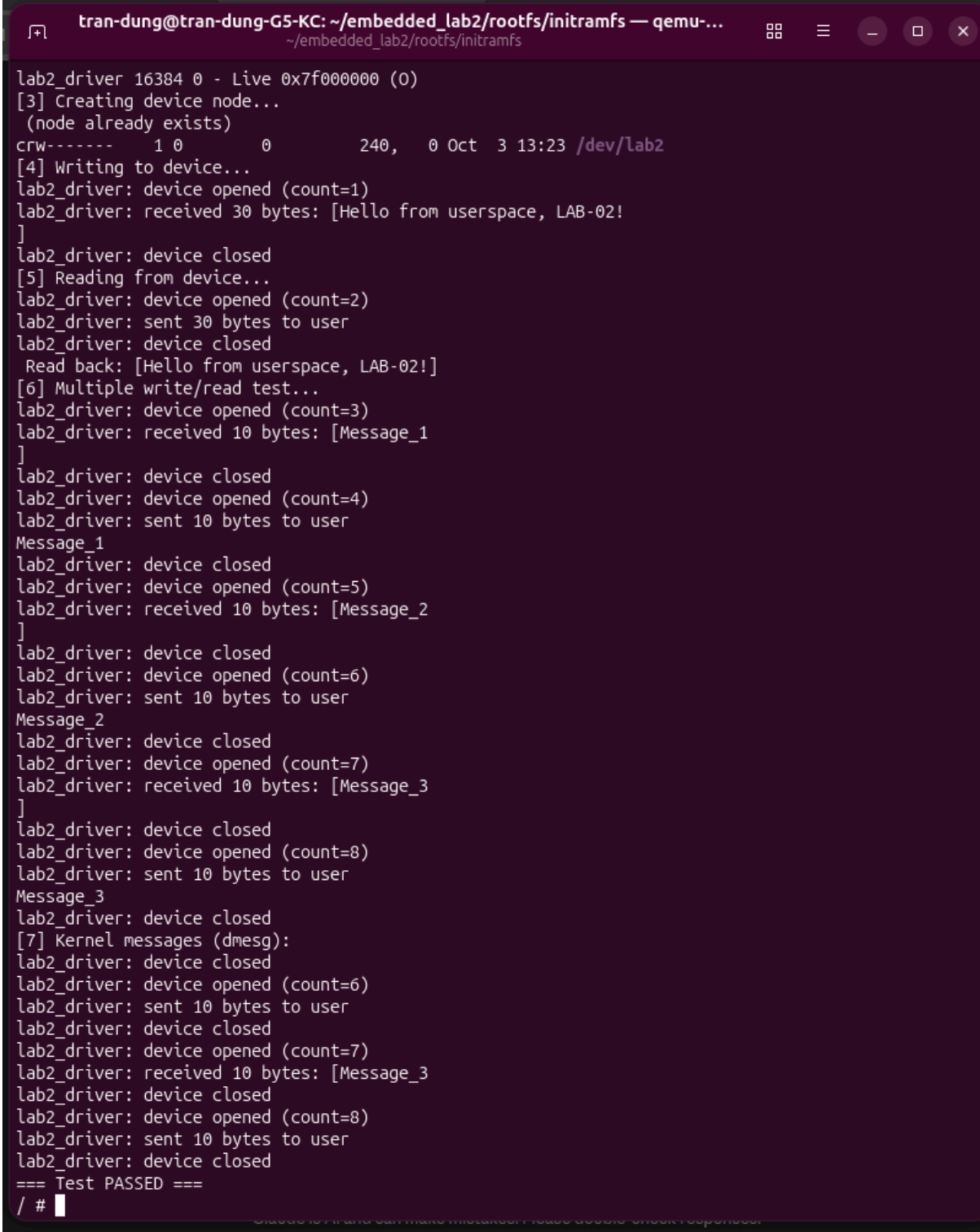
## Mục 3: Phần A

```
tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$ make
make -C /home/tran-dung/embedded_lab1/kernel/linux-5.15 M=/home/tran-dung/embedded_lab2/driver ARCH=arm \
    CROSS_COMPILE=arm-linux-gnueabihf- KCFLAGS="-march=armv7-a -mtune=cortex-a9" modules
CC [M] /home/tran-dung/embedded_lab2/driver/lab2_driver.o
WARNING: Symbol version dump "Module.symvers" is missing.
        Modules may not have dependencies or modversions.
        You may get many unresolved symbol warnings.
MODPOST /home/tran-dung/embedded_lab2/driver/Module.symvers
WARNING: modpost: "device_destroy" [/home/tran-dung/embedded_lab2/driver/lab2_driver.ko] undefined!
WARNING: modpost: "class_destroy" [/home/tran-dung/embedded_lab2/driver/lab2_driver.ko] undefined!
WARNING: modpost: "device_create" [/home/tran-dung/embedded_lab2/driver/lab2_driver.ko] undefined!
WARNING: modpost: "__unregister_chrdev" [/home/tran-dung/embedded_lab2/driver/lab2_driver.ko] undefined!
WARNING: modpost: "__class_create" [/home/tran-dung/embedded_lab2/driver/lab2_driver.ko] undefined!
WARNING: modpost: "__register_chrdev" [/home/tran-dung/embedded_lab2/driver/lab2_driver.ko] undefined!
WARNING: modpost: "memset" [/home/tran-dung/embedded_lab2/driver/lab2_driver.ko] undefined!
WARNING: modpost: "arm_copy_from_user" [/home/tran-dung/embedded_lab2/driver/lab2_driver.ko] undefined!
WARNING: modpost: "warn_slowpath_fmt" [/home/tran-dung/embedded_lab2/driver/lab2_driver.ko] undefined!
WARNING: modpost: "arm_copy_to_user" [/home/tran-dung/embedded_lab2/driver/lab2_driver.ko] undefined!
WARNING: modpost: suppressed 4 unresolved symbol warnings because there were too many)
CC [M] /home/tran-dung/embedded_lab2/driver/lab2_driver.mod.o
LD [M] /home/tran-dung/embedded_lab2/driver/lab2_driver.ko
tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$
```

**Hình A1:** Build kernel module bằng `make` — tạo thành công `lab2_driver.ko` cho kiến trúc ARM.

```
tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$ file lab2_driver.ko
lab2_driver.ko: ELF 32-bit LSB relocatable, ARM, EABI5 version 1 (SYSV), BuildID [sha1]=5612d272002e21137f95114412efc979e4a122e6, with debug_info, not stripped
tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$ modinfo lab2_driver.ko
filename:       /home/tran-dung/embedded_lab2/driver/lab2_driver.ko
version:        1.0
description:    LAB-02: Simple Character Device Driver
author:         Embedded Linux Lab
license:        GPL
srcversion:     4E89036A2D2BA07B7DE24B7
depends:
name:           lab2_driver
vermagic:       5.15.0 SMP mod_unload ARMv7 p2v8
tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$
```

Hình A2: Kết quả `file` và `modinfo` — vermagic khớp kernel 5.15.0, module đúng kiến trúc ARM.



```
tran-dung@tran-dung-G5-KC: ~/embedded_lab2/rootfs/initramfs — qemu-...
~/embedded_lab2/rootfs/initramfs
lab2_driver 16384 0 - Live 0x7f000000 (0)
[3] Creating device node...
(node already exists)
crw----- 1 0 0 240, 0 Oct 3 13:23 /dev/lab2
[4] Writing to device...
lab2_driver: device opened (count=1)
lab2_driver: received 30 bytes: [Hello from userspace, LAB-02!]
]
lab2_driver: device closed
[5] Reading from device...
lab2_driver: device opened (count=2)
lab2_driver: sent 30 bytes to user
lab2_driver: device closed
Read back: [Hello from userspace, LAB-02!]
[6] Multiple write/read test...
lab2_driver: device opened (count=3)
lab2_driver: received 10 bytes: [Message_1
]
lab2_driver: device closed
lab2_driver: device opened (count=4)
lab2_driver: sent 10 bytes to user
Message_1
lab2_driver: device closed
lab2_driver: device opened (count=5)
lab2_driver: received 10 bytes: [Message_2
]
lab2_driver: device closed
lab2_driver: device opened (count=6)
lab2_driver: sent 10 bytes to user
Message_2
lab2_driver: device closed
lab2_driver: device opened (count=7)
lab2_driver: received 10 bytes: [Message_3
]
lab2_driver: device closed
lab2_driver: device opened (count=8)
lab2_driver: sent 10 bytes to user
Message_3
lab2_driver: device closed
[7] Kernel messages (dmesg):
lab2_driver: device closed
lab2_driver: device opened (count=6)
lab2_driver: sent 10 bytes to user
lab2_driver: device closed
lab2_driver: device opened (count=7)
lab2_driver: received 10 bytes: [Message_3
lab2_driver: device closed
lab2_driver: device opened (count=8)
lab2_driver: sent 10 bytes to user
lab2_driver: device closed
=== Test PASSED ===
/ #
```

Hình A3: Chạy `test_driver.sh` trong QEMU — toàn bộ thao tác write/read đơn và lặp lại đều thành công, `dmesg` xác nhận đúng vòng đời driver, kết thúc "Test PASSED".

## Phần B

```

/ # test_procfs.sh
=== Test procfs & sysfs ===
insmod: can't insert '/lib/modules/5.15.0/lab2_driver.ko': File exists
--- /proc/lab2_info (before write) ---
LAB-02 Driver Statistics
=====
Driver name   : lab2
Major number  : 240
Buffer size   : 1024 bytes
Data length   : 10 bytes
Open count    : 8
Last data     : [Message_3
]
lab2_driver: device opened (count=9)
lab2_driver: received 30 bytes: [Embedded Linux Lab2 Test Data
]
lab2_driver: device closed
--- /proc/lab2_info (after write) ---
LAB-02 Driver Statistics
=====
Driver name   : lab2
Major number  : 240
Buffer size   : 1024 bytes
Data length   : 30 bytes
Open count    : 9
Last data     : [Embedded Linux Lab2 Test Data
]
--- sysfs attributes ---
buffer_len    : 30
open_count    : 9
last_data     : Embedded Linux Lab2 Test Data
=== procfs/sysfs Test PASSED ===
/ # █

```

Hình B1: Chạy **test\_procfs.sh** trong QEMU — **/proc/lab2\_info** cập nhật đúng trước/sau khi ghi dữ liệu, các thuộc tính sysfs (**buffer\_len**, **open\_count**, **last\_data**) khớp hoàn toàn với procfs.

```

/ # cat /proc/lab2_info
LAB-02 Driver Statistics
=====
Driver name   : lab2
Major number  : 240
Buffer size   : 1024 bytes
Data length   : 30 bytes
Open count    : 9
Last data     : [Embedded Linux Lab2 Test Data
]
/ # ls /sys/class/lab2_class/lab2/
buffer_len  last_data  power      uevent
dev          open_count subsystem
/ # dmesg | grep lab2
lab2_driver: loading out-of-tree module taints kernel.
lab2_driver: initializing module
lab2_driver: /proc/lab2_info created
lab2_driver: loaded, major=240
lab2_driver: module unloaded
lab2_driver: initializing module
lab2_driver: /proc/lab2_info created
lab2_driver: loaded, major=240
lab2_driver: device opened (count=1)
lab2_driver: received 30 bytes: [Hello from userspace, LAB-02!
lab2_driver: device closed
lab2_driver: device opened (count=2)
lab2_driver: sent 30 bytes to user
lab2_driver: device closed
lab2_driver: device opened (count=3)
lab2_driver: received 10 bytes: [Message_1
lab2_driver: device closed
lab2_driver: device opened (count=4)
lab2_driver: sent 10 bytes to user
lab2_driver: device closed
lab2_driver: device opened (count=5)
lab2_driver: received 10 bytes: [Message_2
lab2_driver: device closed
lab2_driver: device opened (count=6)
lab2_driver: sent 10 bytes to user
lab2_driver: device closed
lab2_driver: device opened (count=7)
lab2_driver: received 10 bytes: [Message_3
lab2_driver: device closed
lab2_driver: device opened (count=8)
lab2_driver: sent 10 bytes to user
lab2_driver: device closed
lab2_driver: device opened (count=9)
lab2_driver: received 30 bytes: [Embedded Linux Lab2 Test Data
lab2_driver: device closed
/ #

```

Hình B2: Kiểm tra trực tiếp **/proc/lab2\_info**, cấu trúc thư mục **/sys/class/lab2\_class/lab2/**, và **dmesg** đầy đủ — thể hiện toàn bộ vòng đời module từ lúc tự động load khi boot đến các thao tác read/write qua cả hai test script.

## Phần C

```

tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$ sudo modprobe nandsim first_id_byte=0x20 second_id_byte=0x35
tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$ sudo modprobe mtdblock
tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$ sudo modprobe jffs2
tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$ cat /proc/mtd
dev:   size  erasesize  name
mtd0: 01000000 00001000 "BIOS"
mtd1: 02000000 00004000 "NAND simulator partition 0"
tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$ ls /dev/mtd*
/dev/mtd0 /dev/mtd0ro /dev/mtd1 /dev/mtd1ro /dev/mtdblock0 /dev/mtdblock1

/dev/mtd:
by-name
tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$

```

**Hình C1:** Kết quả `cat /proc/mtd` và `ls /dev/mtd*` sau khi load `nandsim` — xác định NAND simulator nằm ở `mtd1` (32MB, erasesize 16KB), không phải `mtd0` (đã bị chiếm bởi "BIOS" có sẵn trên host).

```

tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$ sudo mtdinfo /dev/mtd1
mtd1
Name:                NAND simulator partition 0
Type:                nand
Eraseblock size:     16384 bytes, 16.0 KiB
Amount of eraseblocks: 2048 (33554432 bytes, 32.0 MiB)
Minimum input/output unit size: 512 bytes
Sub-page size:       256 bytes
OOB size:            16 bytes
Character device major/minor: 90:2
Bad blocks are allowed: true
Device is writable:  true

```

**Hình C2:** Thông tin chi tiết MTD1 qua `mtdinfo` — eraseblock size 16KB, dung lượng 32MB, ghi được (writable).

```

tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$ sudo flash_erase /dev/mtd1 0 0
Erasing 32768 Kibyte @ 0 -- 100 % complete
tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$

```

**Hình C3:** Xóa toàn bộ NAND flash bằng `flash_erase` trước khi ghi dữ liệu mới — hoàn tất 100%.

```

tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$ sudo mkfs.jffs2 --root=/tmp/jffs2_data --eraseblock=0x4000 --output=/tmp/rootfs.jffs2 --pad --no-cleanmarkers
tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$ ls -lh /tmp/rootfs.jffs2
-rw-r--r-- 1 root root 16K Sep 25 10:19 /tmp/rootfs.jffs2
tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$ ls -lh /tmp/rootfs.jffs2
-rw-r--r-- 1 root root 16K Sep 25 10:19 /tmp/rootfs.jffs2
tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$

```

**Hình C4:** Tạo JFFS2 image từ thư mục mẫu bằng `mkfs.jffs2` với eraseblock 16KB — kết quả `rootfs.jffs2` (16K, đã pad đủ 1 erase block).

```

tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$ ls -la /mnt/jffs2/
total 4
drwxr-xr-x 4 root      root          0 Jan  1  1970 .
drwxr-xr-x 3 root      root        4096 Sep 25 10:22 ..
-rw-rw-r-- 1 tran-dung tran-dung    18 Sep 25 10:19 hello.txt
-rw-rw-r-- 1 tran-dung tran-dung    20 Sep 25 10:19 info.txt
drwxrwxr-x 2 tran-dung tran-dung     0 Sep 25 10:19 logs
tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$ cat /mnt/jffs2/hello.txt
Hello from JFFS2!
tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$ cat /mnt/jffs2/info.txt
Embedded Linux Lab2
tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$ cat /mnt/jffs2/logs/boot.log
Fri Sep 25 10:19:16 AM +07 2026
tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$

```

**Hình C5:** Mount JFFS2 và đọc nội dung các file đã ghi (`hello.txt`, `info.txt`, `logs/boot.log`) — dữ liệu khớp đúng với thư mục mẫu ban đầu.

```

tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$ sudo sh -c 'echo "Written at $(date)" >> /mnt/jffs2/logs/boot.log'
tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$ sudo sh -c 'echo "New runtime file" > /mnt/jffs2/runtime.txt'
tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$ cat /mnt/jffs2/logs/boot.log
Fri Sep 25 10:19:16 AM +07 2026
Written at Fri Sep 25 10:26:42 AM +07 2026
tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$ sudo umount /mnt/jffs2
tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$ sudo mount -t jffs2 /dev/mtdblock1 /mnt/jffs2
tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$ cat /mnt/jffs2/logs/boot.log
Fri Sep 25 10:19:16 AM +07 2026
Written at Fri Sep 25 10:26:42 AM +07 2026
tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$ cat /mnt/jffs2/runtime.txt
New runtime file
tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$

```

**Hình C6:** Kiểm tra tính bền vững (persistence) của JFFS2 — ghi thêm dữ liệu, umount, mount lại, dữ liệu (`boot.log`, `runtime.txt`) vẫn còn nguyên vẹn, chứng minh JFFS2 lưu trữ bền vững trên flash.

```
tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$ sudo flash_erase /dev/mtd1 0 0
Erasing 32768 Kibyte @ 0 -- 100 % complete
tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$ sudo nandwrite -p /dev/mtd1 /tmp/rootfs.jffs2
Writing data to block 0 at offset 0x0
Written 30 blocks containing only 0xff bytes
Those block may be incorrectly treated as empty!
tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$ sudo nanddump -o -l 512 -f /tmp/dump.bin /dev/mtd1
ECC failed: 0
ECC corrected: 0
Number of bad blocks: 0
Number of bbt blocks: 0
Block size 16384, page size 512, OOB size 16
Dumping data starting at 0x00000000 and ending at 0x00000200...
tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$ hexdump -C /tmp/dump.bin | head -20
00000000  85 19 01 e0 31 00 00 00 1d d9 82 42 01 00 00 00 |....1.....B....|
00000010  00 00 00 00 02 00 00 00 27 e8 b5 6a 09 08 00 00 |.....'..j....|
00000020  fe d5 3e a0 15 da 69 f4 68 65 6c 6c 6f 2e 74 78 |..>...i.hello.tx|
00000030  74 ff ff ff 85 19 02 e0 56 00 00 00 09 64 e7 62 |t.....V.....d.b|
00000040  02 00 00 00 01 00 00 00 b4 81 00 00 e8 03 e8 03 |.....|
00000050  12 00 00 00 27 e8 b5 6a 27 e8 b5 6a 27 e8 b5 6a |....'..j'..j'..j|
00000060  00 00 00 00 12 00 00 00 12 00 00 00 00 00 00 00 |.....|
00000070  ee dd 4b b3 4d 2f a0 05 48 65 6c 6c 6f 20 66 72 |..K.M/..Hello fr|
00000080  6f 6d 20 4a 46 46 53 32 21 0a ff ff 85 19 01 e0 |om JFFS2!.....|
00000090  30 00 00 00 78 be 3e fa 01 00 00 00 01 00 00 00 |0...X.>.....|
000000a0  03 00 00 00 2b e8 b5 6a 08 08 00 00 2a 7a e0 7a |....+..j....*z.z|
000000b0  a9 de 79 72 69 6e 66 6f 2e 74 78 74 85 19 02 e0 |..yrinfo.txt....|
000000c0  58 00 00 00 3a 13 38 82 03 00 00 00 01 00 00 00 |X....:8.....|
000000d0  b4 81 00 00 e8 03 e8 03 14 00 00 00 2b e8 b5 6a |.....+.~j|
000000e0  2b e8 b5 6a 2b e8 b5 6a 00 00 00 00 14 00 00 00 |+~j+~j.....|
000000f0  14 00 00 00 00 00 00 00 8a 74 43 dd 5f fb 49 3b |.....tC_.I;|
00000100  45 6d 62 65 64 64 65 64 20 4c 69 6e 75 78 20 4c |Embedded Linux L|
00000110  61 62 32 0a 85 19 01 e0 2c 00 00 00 5f 56 f1 e0 |ab2....._V..|
00000120  01 00 00 00 02 00 00 00 04 00 00 00 34 e8 b5 6a |.....4..j|
00000130  04 04 00 00 0c 42 e7 c1 40 19 cb d1 6c 6f 67 73 |.....B..@...logs|
tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$
```

Claude is AI and can make mistakes. Please double-check responses.

**Hình C7:** Dump raw dữ liệu MTD bằng `nanddump` và xem hex bằng `hexdump` — thấy rõ magic bytes JFFS2 (85 19) ở đầu file và nội dung file (`hello.txt`, "Hello from JFFS2!") đã ghi trong flash.

```
tran-dung@tran-dung-G5-KC: ~/embedded_lab2/driver
~/embedded_lab2/driver

tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$ sudo umount /mnt/jffs2 2>/dev/
null
[sudo: authenticate] Password:
tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$ sudo rmdir nandsim
rmdir: ERROR: Module nandsim is not currently loaded
tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$ ls /dev/mtd* 2>/dev/null || echo 'MTD devices removed'
/dev/mtd0 /dev/mtd0ro

/dev/mtd:
by-name
tran-dung@tran-dung-G5-KC:~/embedded_lab2/driver$
```

**Hình C8:** Dọn dẹp sau khi hoàn tất thử nghiệm — gỡ `nandsim`, xác nhận `/dev/mtd*` chỉ còn thiết bị gốc của hệ thống (BIOS), không còn NAND simulator.

## Phần D

```
tran-dung@tran-dung-G5-KC: ~/embedded_lab2/rootfs/initramfs — qemu-...
~/embedded_lab2/rootfs/initramfs

mmc-pl18x 10005000.mmc: mmc0: PL181 manf 41 rev0 at 0x10005000 irq 33,34 (pio)
10009000.uart: ttyAMA0 at MMIO 0x10009000 (irq = 37, base_baud = 0) is a PL011 rev1
printk: console [ttyAMA0] enabled
1000a000.uart: ttyAMA1 at MMIO 0x1000a000 (irq = 38, base_baud = 0) is a PL011 rev1
1000b000.uart: ttyAMA2 at MMIO 0x1000b000 (irq = 39, base_baud = 0) is a PL011 rev1
1000c000.uart: ttyAMA3 at MMIO 0x1000c000 (irq = 40, base_baud = 0) is a PL011 rev1
rtc-pl031 10017000.rtc: registered as rtc0
rtc-pl031 10017000.rtc: setting system clock to 2026-10-03T13:20:25 UTC (1791033625)
amba 10020000.clcd: Fixing up cyclic dependency with 0-0039
drm-clcd-pl111 10020000.clcd: DVI muxed to daughterboard 1 (core tile) CLCD
drm-clcd-pl111 10020000.clcd: initializing Versatile Express PL111
sii902x 0-0039: supply iovcc not found, using dummy regulator
sii902x 0-0039: supply cvcc12 not found, using dummy regulator
i2c i2c-0: Added multiplexed i2c bus 2
drm-clcd-pl111 1001f000.clcd: assigned reserved memory node vram@4c000000
drm-clcd-pl111 1001f000.clcd: using device-specific reserved memory
drm-clcd-pl111 1001f000.clcd: core tile graphics present
drm-clcd-pl111 1001f000.clcd: this device will be deactivated
drm-clcd-pl111 1001f000.clcd: Versatile Express init failed - -19
drm-clcd-pl111 10020000.clcd: DVI muxed to daughterboard 1 (core tile) CLCD
drm-clcd-pl111 10020000.clcd: initializing Versatile Express PL111
drm-clcd-pl111 10020000.clcd: found bridge on endpoint 0
drm-clcd-pl111 10020000.clcd: Using non-panel bridge
[drm] Initialized pl111 1.0.0 20170317 for 10020000.clcd on minor 0
input: AT Raw Set 2 keyboard as /devices/platform/bus@40000000/bus@40000000:motherboard-bus@40000000/bus@40000000:motherboard-bus@40000000:iofpga@7,00000000/10006000.kmi/serio0/input/input0
Console: switching to colour frame buffer device 128x48
drm-clcd-pl111 10020000.clcd: [drm] fb0: pl111 frame buffer device
ALSA device list:
 #0: ARM AC'97 Interface PL041 rev0 at 0x10004000, irq 32
Freeing unused kernel image (initmem) memory: 1024K
Run /sbin/init as init process
input: ImExPS/2 Generic Explorer Mouse as /devices/platform/bus@40000000/bus@40000000:motherboard-bus@40000000/bus@40000000:motherboard-bus@40000000:iofpga@7,00000000/10007000.kmi/serio1/input/input2
[ LAB-02 ] Starting system initialization...
mount: mounting none on /tmp failed: No such file or directory
[ LAB-02 ] Loading lab2_driver module...
lab2_driver: loading out-of-tree module taints kernel.
lab2_driver: initializing module
lab2_driver: /proc/lab2_info created
lab2_driver: loaded, major=240
[ LAB-02 ] lab2_driver loaded OK
mknod: /dev/lab2: File exists
[ LAB-02 ] /dev/lab2 created (major=240, minor=0)
[ LAB-02 ] System initialization complete!
-> Run test_driver.sh to test character device
-> Run test_procfs.sh to test proc/sysfs interfaces
-> cat /proc/lab2_info to view driver stats

Please press Enter to activate this console. random: fast init done
█
```

Hình D1: Log boot QEMU — **lab2\_driver** được tự động load qua **rcS**, **/proc/lab2\_info** và **/dev/lab2** được tạo tự động, không cần **insmod** thủ công.

```
Please press Enter to activate this console. random: fast init done

/ # lsmod
lab2_driver 16384 0 - Live 0x7f000000 (0)
/ # ls -la /dev/lab2
crw-rw-rw- 1 0 0 240, 0 Oct 3 13:20 /dev/lab2
/ # █
```

Hình D2: Kiểm tra trong QEMU — module **lab2\_driver** đã tự động load (**lsmod**), device node **/dev/lab2** đã tự tạo với major 240 mà không cần thao tác thủ công.